THE AUTOMORPHISM GROUP OF THE $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT COMPANION CALCULATION TO *AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT*

CLAUDE OPUS 4.8

ABSTRACT. The main paper leaves open the identification of $\mathrm{Aut}(\Lambda, +, \star)$, the group of $\mathbb{Z}$-linear bijections of the Leech lattice that preserve the $\mathbb{Z}_3$-symmetric σ-twisted triple-octonion product $\star$. That group has order 36 and is isomorphic to $C_6 \times S_3$: every element is a blockwise octonion automorphism composed with a permutation of the three octonion blocks, the blockwise part running over a cyclic group of order 6 of signed permutations of the octonion basis. It is contained in the Conway group Co_0, and this containment is a theorem rather than an observation: the ambient algebraic group $\mathrm{Aut}(\mathbb{R}^{24}, \star)$ is the compact group $G_2 \times G_2 \times S_3$ of dimension 28, and compactness forces every $\star$-automorphism to be a Euclidean isometry.

1. SETTING

The objects are those of the main paper, restated here only far enough to make this note self-contained.

$\mathbb{O}$ is the real octonion algebra on $\mathbb{R}^8$ with basis $e_0, \dots, e_7$, unit e_0, and the standard Fano triples $(1, 2, 4)$, $(2, 3, 5)$, $(3, 4, 6)$, $(4, 5, 7)$, $(5, 6, 1)$, $(6, 7, 2)$, $(7, 1, 3)$: for each cyclic triple (a, b, c) one has $e_a e_b = +e_c$, for each anticyclic one $e_a e_b = -e_c$, and $e_i^2 = -e_0$ for $i = 1, \dots, 7$. Write N for the norm form and $\mathrm{Re}(x)$ for the e_0-component of x.

$\sigma = (1\ 2)$ is the linear involution of $\mathbb{R}^8$ fixing e_0 and interchanging e_1 and e_2. The σ-twisted octonion product is

$$x \cdot_s y \ := \ \sigma\big(\sigma(x)\, \sigma(y)\big),$$

which is again an octonion product on $\mathbb{R}^8$, and σ is by construction an algebra isomorphism $(\mathbb{O}, \cdot) \to (\mathbb{O}, \cdot_s)$.

$L = D_8^+$ is the E_8 lattice realized in octonion coordinates, $s = \frac{1}{2}(-e_0 + e_1 + \cdots + e_7)$, $\bar{s} = \frac{1}{2}(-e_0 - e_1 - \cdots - e_7)$, and $Ls = \{\ell \cdot s : \ell \in L\}$, $L\bar{s} = \{\ell \cdot \bar{s} : \ell \in L\}$, both formed with the *untwisted* octonion product. Wilson's description of the Leech lattice inside L^3 is

$$\Lambda \ = \ \big\{(x, y, z) \in L^3 \ : \ x + y,\ x + z,\ y + z \in L\bar{s};\ \ x + y + z \in Ls\big\}. \tag{1}$$

On $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ the product of the main paper is

$$(x, y, z) \star (x', y', z') \ = \ \big(\, x \cdot_s x' + z \cdot_s y' + y \cdot_s z',\ \ y \cdot_s y' + x \cdot_s z' + z \cdot_s x',\ \ z \cdot_s z' + y \cdot_s x' + x \cdot_s y' \,\big), \tag{2}$$

and the main paper's Theorem 1.1 is $\Lambda \star \Lambda \subseteq \Lambda$.

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This note was computed at the direction of Jens Köplinger and is companion material to the main paper *An order on the Leech lattice from a $\mathbb{Z}_3$-symmetric triple-octonion product* (v6). It is not part of that paper and is not in its referee scope; the author of the main paper will decide how, and how much of, this material is worked into it. Every nontrivial statement below is either proved here or reproduced by a named script in the project repository. Two independent computational passes were run: a first pass that produced the result, and a second pass that rebuilt the lattice, the product and the generators from the paper's definitions without reusing any code from the first. Where the two passes disagreed, the disagreement is recorded in §7.

Throughout, blocks are indexed by $\mathbb{Z}/3 = \{0, 1, 2\}$ and a vector of $\mathbb{R}^{24}$ is written $u = (u_0, u_1, u_2)$ with $u_\alpha \in \mathbb{O}$. The object under study is

$$\mathrm{Aut}(\Lambda, +, \star) \ := \ \big\{ g : \Lambda \to \Lambda \text{ a } \mathbb{Z}\text{-linear bijection with } g(u \star v) = g(u) \star g(v) \text{ for all } u, v \in \Lambda \big\}.$$

The ambient group here is $\mathrm{GL}_{24}(\mathbb{Z})$ and not Co_0: that every such g is in fact an isometry is one of the things settled below (§6), not something assumed.

Two conventions should be flagged. First, $\star$ is bilinear and Λ-membership is $\mathbb{Z}$-linear, so a check carried out on all $24 \times 24 = 576$ pairs of basis vectors is a *proof* for all of $\mathbb{R}^{24}$, not a sample; every "exhaustive basis check" below is of this kind. Second, Λ as realized by (1) has minimal norm 8, Gram determinant 2^{24} and index 4096 in L^3: it is a copy of $\sqrt{2}\,\Lambda_{\mathrm{Leech}}$ rather than the unimodular Leech lattice. Nothing below depends on the scale (Co_0 is scale invariant), but no claim of unimodularity should be attached to this realization.

2. The routing identity and the ideal decomposition

The nine octonion blocks in (2) are routed by a single congruence.

Lemma 1 (Routing). *For $u, v \in \mathbb{R}^{24}$ and $\gamma \in \mathbb{Z}/3$,*

$$(u \star v)_\gamma \ = \sum_{\substack{\alpha,\beta \in \mathbb{Z}/3 \\ \alpha+\beta \equiv 2\gamma}} u_\alpha \cdot_s v_\beta.$$

Proof. Read off (2) with $(u_0, u_1, u_2) = (x, y, z)$. The first block collects the index pairs $(0,0), (2,1), (1,2)$, all with $\alpha + \beta \equiv 0 = 2 \cdot 0$; the second collects $(1,1), (0,2), (2,0)$, all with $\alpha + \beta \equiv 2 = 2 \cdot 1$; the third collects $(2,2), (1,0), (0,1)$, all with $\alpha + \beta \equiv 1 \equiv 2 \cdot 2$. ☐

Since 2 is invertible modulo 3, each of the nine pairs (α, β) occurs in exactly one block, namely $\gamma \equiv 2(\alpha + \beta)$. Two consequences are immediate.

Proposition 2 (Ideal decomposition). *Let*

$$D \ := \ \{(a, a, a) : a \in \mathbb{O}\}, \qquad T \ := \ \{(p, q, r) \in \mathbb{O}^3 : p + q + r = 0\},$$

and let $\Sigma(u) := u_0 + u_1 + u_2$. Then

1. *$\Sigma(u \star v) = \Sigma(u) \cdot_s \Sigma(v)$, so Σ is a surjective algebra homomorphism $(\mathbb{R}^{24}, \star) \to (\mathbb{O}, \cdot_s)$;*
2. *$(a, a, a) \star v = \big(a \cdot_s \Sigma(v)\big)^{\mathrm{diag}}$ and $v \star (a, a, a) = \big(\Sigma(v) \cdot_s a\big)^{\mathrm{diag}}$, where $w^{\mathrm{diag}} := (w, w, w)$;*
3. *D and $T = \ker \Sigma$ are two-sided ideals, $D \star T = T \star D = 0$, and $\mathbb{R}^{24} = D \oplus T$;*
4. *$\varphi(a) := \frac{1}{3}(a, a, a)$ is an algebra isomorphism $(\mathbb{O}, \cdot_s) \to (D, \star)$; equivalently $(a, a, a) \star (b, b, b) = 3\,(a \cdot_s b,\ a \cdot_s b,\ a \cdot_s b)$;*
5. *$D \perp T$ for the standard Euclidean form of $\mathbb{R}^{24}$.*

Proof. (1) Summing Lemma 1 over γ collects each pair (α, β) exactly once, giving $\sum_{\alpha,\beta} u_\alpha \cdot_s v_\beta = \Sigma(u) \cdot_s \Sigma(v)$ by bilinearity. Surjectivity is clear from $\Sigma(a, 0, 0) = a$. (2) With $u = (a, a, a)$ the inner sum of Lemma 1 is $\sum_\beta a \cdot_s v_\beta = a \cdot_s \Sigma(v)$ for every γ, independent of γ; the right-multiplication case is the same computation. (3) T is a two-sided ideal because it is the kernel of the homomorphism Σ; D is a two-sided ideal by (2). If $v \in T$ then $\Sigma(v) = 0$, so (2) gives $D \star T = T \star D = 0$. Finally $\Sigma|_D : (a, a, a) \mapsto 3a$ is bijective onto $\mathbb{O}$ and $T = \ker \Sigma$, so $\mathbb{R}^{24} = D \oplus T$. (4) Immediate from (2) with $v = (b, b, b)$. (5) $\langle (a, a, a), (p, q, r) \rangle = \langle a, p + q + r \rangle = 0$. ☐

All five items were additionally checked by exhaustive basis computation in exact rational arithmetic, and re-checked against an independently coded nine-term expansion of $\star$ (`python_project/src/verify_ideal_decomposition.py`).

Remark 3. The same script verifies that $\sigma^{\oplus 3} : (x, y, z) \mapsto (\sigma x, \sigma y, \sigma z)$ is an algebra isomorphism from the σ-twisted triple product onto the untwisted one. The twist is invisible to the algebra: only the lattice Λ sees it. Consequently everything in §2–§4 could be stated without σ; σ re-enters in §5, where the lattice conditions do.

Every $\star$-automorphism preserves D and T. Because $D \star T = T \star D = 0$, the two projections π_D, π_T associated with $\mathbb{R}^{24} = D \oplus T$ are algebra homomorphisms: for $u = u_D + u_T$ and $v = v_D + v_T$, $u \star v = u_D \star v_D + u_T \star v_T$ with the first term in D and the second in T, so $\pi_T(u \star v) = \pi_T(u) \star \pi_T(v)$ and likewise for π_D.

Lemma 4. *$(D, \star)$ and $(T, \star)$ are simple algebras.*

Proof and computation. $D \cong (\mathbb{O}, \cdot_s)$ by Proposition 2(4), and the real octonions form a division algebra: if $I \subseteq \mathbb{O}$ is a nonzero ideal and $0 \neq a \in I$, then right multiplication R_a is invertible and $R_a(\mathbb{O}) \subseteq I$, so $I = \mathbb{O}$. This is a proof.

For T the statement is established by exact linear algebra: the multiplication algebra of T, that is the unital associative subalgebra of $\operatorname{End}(T)$ generated by all left and right $\star$-multiplications by elements of T, has dimension $256 = \dim \operatorname{End}(T)$, so it is all of $\operatorname{End}(T) \cong M_{16}(\mathbb{R})$. Any ideal of T is invariant under the multiplication algebra, hence under all of $\operatorname{End}(T)$, hence is 0 or T. The dimension is a rank of an explicit rational matrix; it was computed exactly, with a full-rank certificate modulo a prime as a rigorous lower bound. (Second pass, independent rebuild; the same computation gives dimension $64 = \dim \operatorname{End}(D)$ for D, reproving the first paragraph.) ☐

Proposition 5. *Every $g \in \operatorname{Aut}(\mathbb{R}^{24}, \star)$ satisfies $g(D) = D$ and $g(T) = T$, and consequently*

$$\operatorname{Aut}(\mathbb{R}^{24}, \star) \;=\; \operatorname{Aut}(D, \star) \times \operatorname{Aut}(T, \star),$$

the two restrictions being independently choosable.

Proof. $g(D)$ is an ideal of $\mathbb{R}^{24}$ of dimension 8. Applying the surjective algebra homomorphism π_T, the image $\pi_T(g(D))$ is an ideal of T of dimension at most $8 < 16$. By Lemma 4 it is therefore 0, so $g(D) \subseteq \ker \pi_T = D$, and $g(D) = D$ by bijectivity.

Next, $T = \operatorname{Ann}(D) := \{u : u \star D = D \star u = 0\}$. Indeed $T \subseteq \operatorname{Ann}(D)$ by Proposition 2(3); conversely, write $u = d + t$ with $d \in D$, $t \in T$, and note that $u \star (a, a, a) = d \star (a, a, a)$ for every a, so $u \in \operatorname{Ann}(D)$ forces d to annihilate the division algebra $(D, \star) \cong (\mathbb{O}, \cdot_s)$, whence $d = 0$. (Confirmed as an exact nullspace computation: $\dim \operatorname{Ann}(D) = 16$ and $\dim \operatorname{Ann}(T) = 8$.) Since $g(D) = D$ is now known, $g(T) = g(\operatorname{Ann}(D)) = \operatorname{Ann}(g(D)) = \operatorname{Ann}(D) = T$.

Finally, since $D \star T = T \star D = 0$, any pair of algebra automorphisms of D and of T assembles to an automorphism of $\mathbb{R}^{24}$, and by the above every automorphism arises this way. ☐

Remark 6 (A correction, recorded). The first computational pass justified Proposition 5 by "$\operatorname{Ann}(T) = D$ and $\operatorname{Ann}(D) = T$; annihilators are intrinsic, so every automorphism preserves D and T". As written that is circular: $g(\operatorname{Ann}(D)) = \operatorname{Ann}(g(D))$ yields $g(T) = T$ only *after* $g(D) = D$ is known, and the annihilator relations alone do not establish that D is intrinsic. The simplicity argument above is the non-circular replacement and is what the second pass verified. The conclusion is unaffected. The docstring of `python_project/src/verify_star_algebra_structure.py` still carries the circular wording and should be corrected.

3. The C_3-Fourier normal form

The routing identity diagonalizes under the discrete Fourier transform of the block index. Let ζ be a primitive cube root of unity and, for $u \in \mathbb{R}^{24} \otimes \mathbb{C}$, put

$$\hat{u}_j \;:=\; \sum_{\alpha \in \mathbb{Z}/3} \zeta^{\,j\alpha}\, u_\alpha \;\in\; \mathbb{O}_{\mathbb{C}}, \qquad j \in \mathbb{Z}/3.$$

Proposition 7. *For all u, v,*

$$\widehat{(u\star v)}_k \;=\; \hat u_{2k}\cdot_s \hat v_{2k}, \qquad k\in\mathbb{Z}/3.$$

Proof. By Lemma 1, the block γ receiving the pair (α,β) is $\gamma\equiv 2(\alpha+\beta)$. Hence

$$\widehat{(u\star v)}_k=\sum_\gamma \zeta^{k\gamma}(u\star v)_\gamma=\sum_{\alpha,\beta}\zeta^{2k(\alpha+\beta)}\,u_\alpha\cdot_s v_\beta=\Big(\sum_\alpha \zeta^{2k\alpha}u_\alpha\Big)\cdot_s\Big(\sum_\beta \zeta^{2k\beta}v_\beta\Big)=\hat u_{2k}\cdot_s \hat v_{2k}. \quad \square$$

The identity was also verified exactly on all 576 basis pairs over $\mathbb{Q}(\zeta)$, elements being carried as pairs $p+q\zeta$ with $\zeta^2=-1-\zeta$ (`python_project/src/verify_star_algebra_structure.py`); the second pass verified the equivalent statement in its own coordinates.

The case $k=0$ is Proposition 2(1). The cases $k=1,2$ say that on $T\otimes\mathbb{C}$, with coordinates $(\hat u_1,\hat u_2)\in\mathbb{O}_\mathbb{C}\oplus\mathbb{O}_\mathbb{C}$, the product is the *swap-twisted* algebra

$$S:\qquad (a_1,a_2)\cdot(b_1,b_2)\;=\;(a_2\cdot_s b_2,\; a_1\cdot_s b_1)\qquad \text{on } \mathbb{O}_\mathbb{C}\oplus\mathbb{O}_\mathbb{C}, \tag{3}$$

and that the real structure on S is $(a_1,a_2)\mapsto(\overline{a_2},\overline{a_1})$, since $\hat u_2=\overline{\hat u_1}$ for real u.

This is the structural engine of the calculation: the 16-dimensional piece T, which is where all the interesting automorphisms live, is a real form of a pair of complexified octonion algebras glued by a swap.

4. The automorphism group of the ambient algebra

For $A,B\in\mathrm{Aut}(\mathbb{O},\cdot_s)$ define $h_{A,B}:=P\otimes A+Q\otimes B$, where P is the orthogonal projection of $\mathbb{R}^{24}$ onto D and $Q=I-P$ the projection onto T; concretely

$$h_{A,B}(u)_\alpha\;=\;B\,u_\alpha\;+\;(A-B)\,\tfrac{1}{3}\,(u_0+u_1+u_2). \tag{4}$$

For $\pi\in S_3$ let $\rho_\pi(u)_\alpha:=u_{\pi^{-1}(\alpha)}$ be the corresponding permutation of the three blocks.

Theorem 8.

$$\mathrm{Aut}(\mathbb{R}^{24},\star)\;=\;\{\,h_{A,B}\,\rho_\pi\;:\;A,B\in\mathrm{Aut}(\mathbb{O},\cdot_s),\;\pi\in S_3\,\}\;\cong\;G_2\times G_2\times S_3,$$

a direct *product, where G_2 denotes the compact real form. The group is compact of dimension* 28*; $h_{A,B}$ and ρ_π commute.*

The proof has three parts.

(a) The D factor. $\mathrm{Aut}(D,\star)=\mathrm{Aut}(\mathbb{O},\cdot_s)$ by Proposition 2(4), and $\mathrm{Aut}(\mathbb{O},\cdot_s)$ is the compact real form of G_2 (classical; see also §6). The rescaling by 3 does not affect the automorphism group.

(b) The identity component of the T factor. Exact nullspace computations give $\dim\mathrm{Der}(\mathbb{O},\cdot_s)=14$ and $\dim\mathrm{Der}(\mathbb{R}^{24},\star)=28$: 28 explicit derivations of the form $P\otimes X+Q\otimes Y$ with $X,Y\in\mathrm{Der}(\mathbb{O},\cdot_s)$ are exhibited and shown to be linearly independent over $\mathbb{Q}$, giving dim ≥ 28, and a rank bound modulo a prime (full rank over $\mathbb{F}_p$ is a rigorous lower bound for the rank over $\mathbb{Q}$, hence the mod-p nullity of the derivation system is a rigorous upper bound for its rank over $\mathbb{Q}$) gives dim ≤ 28. Hence $\mathrm{Der}(\mathbb{R}^{24},\star)=\mathfrak{g}_2\oplus\mathfrak{g}_2$; the second pass additionally computed the Killing form of $\mathrm{Der}(\mathbb{O},\cdot_s)$ and found it negative definite, so both summands are of compact type. In particular $\mathrm{Aut}(T,\star)^0$ is the blockwise compact G_2. (`verify_star_algebra_structure.py`; second pass independently.)

(c) The component group of the T factor. Two independent derivations agree.

Route 1 (over $\mathbb{C}$, via (3)*).* In S, for $w=(a_1,a_2)$ the left multiplication L_w has rank 4 exactly when one component vanishes and the other is a nonzero null vector of $\mathbb{O}_\mathbb{C}$. An automorphism preserves that rank condition, so it permutes the two irreducible punctured null cones N_1,N_2 and hence their spans V_1,V_2. The component-preserving automorphisms solve out exactly to $(\phi_1,\phi_2)=(c^2\theta,\,c\theta)$ with $\theta\in\mathrm{Aut}(\mathbb{O}_\mathbb{C})$ and $c^3=1$; the swap $\tau(a_1,a_2)=(a_2,a_1)$ is an automorphism, fixes θ and inverts c. Hence $\mathrm{Aut}(S) = G_2(\mathbb{C}) \times S_3$. Passing to real points, the real structure $(a_1, a_2) \mapsto (\overline{a_2}, \overline{a_1})$ forces θ to be fixed by conjugation, that is θ lies in the compact G_2, while c is unconstrained among the cube roots of unity. So $\mathrm{Aut}(T, \star)(\mathbb{R}) = G_2 \times S_3$, the G_2 acting blockwise and the S_3 being the block permutations restricted to T.

Route 2 (finite and machine-checkable). $\mathrm{Aut}(G_2) = \mathrm{Inn}(G_2)$ and $Z(G_2) = 1$, so the component group of $\mathrm{Aut}(T, \star)$ is the centralizer of the identity component $\mathrm{Aut}(T, \star)^0$ inside $\mathrm{Aut}(T, \star)$. The commutant of the blockwise $\mathfrak{g}_2$ inside $\mathrm{End}(T)$ was computed exactly and is 8-dimensional ($\cong M_2 \oplus M_2$); imposing the automorphism condition on it yields 32 distinct polynomial equations in 8 unknowns, solved exactly, with exactly 6 invertible solutions. They are closed under composition and have orders $1, 2, 2, 2, 3, 3$, hence form S_3. There are no further components.

Route 2 was carried out in the second, independent pass and does not share a step with Route 1. Route 1 is the derivation recorded in `verify_star_algebra_structure.py`. The two agree.

Assembly. By Proposition 5, $\mathrm{Aut}(\mathbb{R}^{24}, \star) = \mathrm{Aut}(D, \star) \times \mathrm{Aut}(T, \star) = G_2 \times (G_2 \times S_3)$. Writing an element as $h_{A,B}\rho_\pi$ uses A for the action on D and B for the blockwise action on T; the block permutations act trivially on D and commute with blockwise maps, so the product is direct and $h_{A,B}\rho_\pi = \rho_\pi h_{A,B}$. That every $h_{A,B}$ and every ρ_π *is* a $\star$-automorphism was checked exhaustively on all 576 basis pairs. □

Remark 9 (What deserves a human reading). Everything in Theorem 8 is either an exact finite computation or elementary, with one exception: the classification of $\mathrm{Aut}(S)$ over $\mathbb{C}$ in Route 1, specifically the rank-of-L_w / null-cone step. It is the only genuinely non-mechanical argument in this note. The second pass deliberately did *not* reproduce it, taking Route 2 instead, and reached the same group. If Route 1 is printed anywhere, it should be re-read by hand; Route 2 is the safer thing to publish.

5. The lattice stabilizer

Since Λ spans $\mathbb{R}^{24}$, a $\mathbb{Z}$-linear bijection of Λ extends uniquely to $\mathbb{R}^{24}$, and $\star$-multiplicativity on Λ extends by bilinearity to all of $\mathbb{R}^{24}$. Hence

$$\mathrm{Aut}(\Lambda, +, \star) \;=\; \{\, g \in \mathrm{Aut}(\mathbb{R}^{24}, \star) : g(\Lambda) = \Lambda \,\} \;=\; \{\, h_{A,B}\, \rho_\pi \;:\; h_{A,B}(\Lambda) = \Lambda,\ \pi \in S_3 \,\},$$

the last equality because every ρ_π preserves Λ: Wilson's three conditions in (1) are symmetric in x, y, z. So the whole problem is the determination of

$$K \;:=\; \big\{(A, B) \in \mathrm{Aut}(\mathbb{O}, \cdot_s)^2 \;:\; h_{A,B}(\Lambda) = \Lambda\big\}, \qquad \mathrm{Aut}(\Lambda, +, \star) \;\cong\; K \times S_3.$$

5.1. **The two lattice slices.**

Lemma 10. *$\Lambda \cap T = \{(p, q, r) \in (L\bar{s})^3 : p + q + r = 0\}$, and $\Lambda \cap D = \{(a, a, a) : a \in Ls\}$.*

Proof of the first, computation of the second. If $(p, q, r) \in \Lambda$ with $p+q+r = 0$, then $p+q = -r \in L\bar{s}$ and likewise for the other two pairwise sums, and $L\bar{s} \subseteq L$, so $p, q, r \in L\bar{s}$; the condition $p + q + r = 0 \in Ls$ is automatic. The converse is the same reading. This is a proof.

For the second slice, $(a, a, a) \in \Lambda$ says $a \in L$, $2a \in L\bar{s}$ and $3a \in Ls$. All three conditions are constant on the classes of $L/2L$ (because $2L \subseteq L\bar{s}$ and $2L \subseteq Ls$, both verified), so they may be tested on the 256 classes; exactly 16 survive, and they are precisely the classes of Ls, which has index 16 in L. The inclusion $\supseteq$ is immediate ($a \in Ls$ gives $3a \in Ls$ and $2a \in 2L \subseteq L\bar{s}$); the reverse inclusion is the computation. (`python_project/src/verify_aut_lambda_star.py`, reproduced in the second pass from an independently built $\mathbb{Z}$-basis of Λ.) □

Corollary 11 (Finiteness). *If $h_{A,B}\rho_\pi \in \mathrm{Aut}(\Lambda, +, \star)$ then $A(Ls) = Ls$ and $B(L\bar{s}) = L\bar{s}$. Both stabilizers are finite, so $\mathrm{Aut}(\Lambda, +, \star)$ is finite.*

Proof. $h_{A,B}$ acts on D as $(a,a,a) \mapsto (Aa, Aa, Aa)$ and preserves D and T; block permutations act trivially on D. By Lemma 10, preserving $\Lambda \cap D$ forces $A(Ls) = Ls$. On T, $h_{A,B}$ acts blockwise by B; every $p \in L\bar{s}$ occurs as the first component of $(p, -p, 0) \in \Lambda \cap T$, so preserving $\Lambda \cap T$ forces $B(L\bar{s}) = L\bar{s}$. A lattice stabilizer inside the orthogonal group is finite. (Alternatively: $\mathrm{Aut}(\mathbb{R}^{24}, \star)$ is compact by Theorem 8 and $\mathrm{Aut}(\Lambda, +, \star)$ is a discrete subgroup of it.) □

5.2. **The three octonion-automorphism stabilizers.** Corollary 11 reduces the search to the two finite groups

$$\mathrm{Stab}(Ls) := \mathrm{Aut}(\mathbb{O}, \cdot_s) \cap \{A : A(Ls) = Ls\}, \qquad \mathrm{Stab}(L\bar{s}) := \mathrm{Aut}(\mathbb{O}, \cdot_s) \cap \{B : B(L\bar{s}) = L\bar{s}\},$$

and it is convenient to have $\mathrm{Stab}(L)$ as well. All three were enumerated completely, in exact arithmetic.

Group	Order	Structure (GAP)
$\mathrm{Stab}(L)$	1344	$(C_2 \times C_2 \times C_2).\mathrm{PSL}(3,2)$
$\mathrm{Stab}(Ls)$	168	$(C_2 \times C_2 \times C_2){:}(C_7{:}C_3)$
$\mathrm{Stab}(L\bar{s})$	12096	$\mathrm{PSU}(3,3){:}C_2 = U_3(3).2 = G_2(2)$
$\mathrm{Stab}(L) \cap \mathrm{Stab}(Ls) \cap \mathrm{Stab}(L\bar{s})$	6	C_6

The identifications are GAP's, from 8×8 generators exported by the Python enumeration (`octonion_stabilisers.g`); the 12096 was additionally confirmed isomorphic to the ATLAS group $U_3(3).2$.

Completeness of the enumerations rests on the following. An automorphism of $\mathbb{O}$ fixes e_0, is orthogonal (§6), permutes the minimal vectors of any lattice it stabilizes, and is determined by the images of *three* elements that generate $\mathbb{O}$ as an algebra. Three, not two: by Artin's theorem any two octonions generate an associative, hence at most quaternionic, subalgebra, so a search over the images of two generators undercounts the constraints and returns a much larger set. A backtracking search over ordered generating triples drawn from the minimal-vector shell, with Gram-matrix filters, is therefore exhaustive.

The enumerations were checked in three further ways. (i) Re-running the backtracking search from a different first generator, which produces a different generating triple, different Gram filters and a different candidate list, returns the identical *set* of matrices in all three cases (`python_project/src/verify_aut_octonion_crosscheck.py`). (ii) For $\mathrm{Stab}(L)$ an assumption-free brute force over all $2^7 \cdot 7! = 645{,}120$ signed permutations of $e_1, \dots, e_7$ returns the same 1344 matrices. This is legitimate because every element of $\mathrm{Stab}(L)$ *is* such a signed permutation: $e_0 \pm e_j$ are roots of L and are the only roots of L of the form $e_0 +$(imaginary unit), so $A(e_0 \pm e_j) = e_0 \pm A(e_j)$ forces $A(e_j) = \pm e_k$. (The forcing argument uses roots of norm 2 and does not apply to Ls or $L\bar{s}$, whose minimal vectors have norm 4; and indeed those two stabilizers do contain non-monomial elements.) (iii) The second pass re-enumerated all three from scratch, by its own shell and triple search, and obtained the same three orders.

Remark 12 (A prior expectation refuted). It is natural to expect the order-12096 group $G_2(2)$ to appear as the stabilizer of the integral-octonion order L. In the paper's normalization it does not. L has minimal norm 2 and does not contain e_0, so it is not a unital maximal order in these coordinates; its octonion-automorphism stabilizer has order 1344 and consists entirely of monomial signed permutations. It is $L\bar{s}$, not L, whose stabilizer is the full $G_2(2)$. The asymmetry between Ls (order 168) and $L\bar{s}$ (order 12096) is the sharpest single fact in this computation.

5.3. **The pair set K and the group.**

Theorem 13. $|K| = 6$*, every element of* K *has* $A = B$*, and*

$$\mathrm{Aut}(\Lambda, +, \star) \;\cong\; C_6 \times S_3, \qquad \big|\mathrm{Aut}(\Lambda, +, \star)\big| \;=\; 36.$$

The diagonal part of K is identified by a proof: when $A = B$, $h_{A,A}$ is just the blockwise map $(x, y, z) \mapsto (Ax, Ay, Az)$, and since A is an octonion-algebra automorphism it commutes with the sublattice conditions of (1) in the sense that $h_{A,A}(\Lambda) = \Lambda$ holds if and only if A stabilizes each of L, $L\bar{s}$ and Ls. So the diagonal part of K is exactly the triple intersection $\mathrm{Stab}(L)\cap\mathrm{Stab}(Ls)\cap\mathrm{Stab}(L\bar{s})$, which the enumeration above gives as a C_6. What is *computed* is that K has no other elements: no pair with $A \neq B$ survives. The lattice kills every genuinely asymmetric pair, even though the ambient algebra permits them freely (Theorem 8 has two independent G_2 factors).

The search over all $168 \times 12096 = 2{,}032{,}128$ candidate pairs was done by two independent methods, which return the identical 6 pairs (`python_project/src/verify_aut_lambda_star.py`):

M1. *The mod-3 glue code.* $\Lambda_0 := (\Lambda \cap D) \oplus (\Lambda \cap T)$ has index $[\Lambda : \Lambda_0] = 6561 = 3^8$ in Λ, and the quotient Λ/Λ_0 is an 8-dimensional $\mathbb{F}_3$-subspace of $\mathbb{F}_3^{24}$. Any $h_{A,B}$ with $A \in \mathrm{Stab}(Ls)$, $B \in \mathrm{Stab}(L\bar{s})$ preserves Λ_0 automatically, so $h_{A,B}(\Lambda) = \Lambda$ if and only if the induced $\mathbb{F}_3$-linear map $\mathrm{diag}(M_A, M_B, M_B)$ preserves the glue code. Matching the 168 classes against the 12096 classes gives exactly 6 pairs.

M2. *Direct integrality.* With no reference to Λ_0, no lattice quotient and no $\mathbb{F}_3$ linear algebra: the matrix of $h_{A,B}$ in a Hermite $\mathbb{Z}$-basis of Λ must be integral, and that condition, tested modulo the common denominator 96 over all 2,032,128 pairs, returns the same 6.

The second pass repeated the search in its own coordinates (hashing the Λ-basis integrality condition over all 2,032,128 pairs) and again got exactly the same 6 pairs, all with $A = B$, forming a C_6 of monomial signed permutations fixing e_0.

5.4. **Generators.** Three generators suffice. Written intrinsically:

(1) **The blockwise C_6 generator**, of order 6: apply A_6 to each of the three octonion blocks, where A_6 is the signed permutation of the octonion basis fixing e_0 and acting by

$$e_1 \mapsto -e_5, \quad e_2 \mapsto -e_7, \quad e_3 \mapsto +e_2, \quad e_4 \mapsto -e_4, \quad e_5 \mapsto +e_6, \quad e_6 \mapsto +e_1, \quad e_7 \mapsto -e_3.$$

A_6 is an automorphism of $(\mathbb{O}, \cdot_s)$ and stabilizes L, Ls and $L\bar{s}$. Its underlying permutation is $(1\,5\,6)(2\,7\,3)$; its cube is the sign flip $e_1, e_4, e_5, e_6 \mapsto -e_1, -e_4, -e_5, -e_6$, and its square is a monomial automorphism of order 3. All six elements of K are monomial signed permutations fixing e_0.

(2) **The block 3-cycle**, of order 3: $(x, y, z) \mapsto (y, z, x)$.

(3) **The block transposition**, of order 2: $(x, y, z) \mapsto (y, x, z)$.

Generators (2) and (3) generate the S_3 of block permutations. Under the Fourier normal form of §3 they have a clean reading: the block 3-cycle is the cube-root-of-unity scaling $(\hat{u}_1, \hat{u}_2) \mapsto (\zeta^2\hat{u}_1, \zeta\hat{u}_2)$, and the block transposition is the Fourier conjugation $\hat{u}_1 \leftrightarrow \hat{u}_2$. That A_6 is not an automorphism of the *untwisted* octonion product was checked in the second pass; the C_6 is a feature of $\cdot_s$.

5.5. **Structure.** The three generators, converted to 24×24 integer matrices in a Hermite $\mathbb{Z}$-basis of Λ, are exported to `aut_lambda_star_gens.g`. GAP, running its own Schreier–Sims machinery and knowing nothing of the Python enumeration, reports (`aut_lambda_star.g`):

$\lvert G\rvert$	36
StructureDescription(G)	`C6 x S3`
abelian	no
center	C_6
derived subgroup	C_3
$-I_{24} \in G$	no
conjugacy classes	18
element orders	1 (×1), 2 (×7), 3 (×8), 6 (×20)

The second pass rebuilt the three generators from their intrinsic description above, in *standard* $\mathbb{R}^{24}$ coordinates rather than in the Hermite basis, closed the group by hand (order 36), and ran GAP again on its own matrices, obtaining the same order, the same structure description, the same center and derived subgroup, and the same exclusion of $-I_{24}$.

All 36 elements were also verified one at a time, in exact arithmetic and twice independently: each preserves Λ, each satisfies $g(u \star v) = g(u) \star g(v)$ on all 576 basis pairs (hence, by bilinearity, everywhere), each preserves the Leech inner product, and the 36 are pairwise distinct linear maps.

That $-I_{24} \notin \mathrm{Aut}(\Lambda, +, \star)$ is not a surprise: negation is an isometry but is not multiplicative for a bilinear product. It is the reason the containment in Co_0 of §6 is strict, and it is already noted in the main paper.

Remark 14 (A mislabeled datum, recorded). The first pass reported the element orders as "$1, 2, 2, 2, 3, 3, 3, 3, 3, 6(\times 9)$". That is the multiset of orders of the 18 *conjugacy class representatives*, which is what the GAP script prints, and not the element-order multiset of the group. The correct element-order distribution is the one tabulated above. Both are consistent with $C_6 \times S_3$, but the former must not be copied into the paper as an element-order list.

6. Containment in Co_0

Theorem 15. $\mathrm{Aut}(\mathbb{R}^{24}, \star) \subseteq O(24)$*, and consequently* $\mathrm{Aut}(\Lambda, +, \star) \subseteq \mathrm{Co}_0$.

Proof. Let A be an algebra automorphism of the real octonions. The trace $t(x) := \frac{1}{8}\,\mathrm{tr}(L_x)$ and hence the conjugation $\bar{x} = t(x)\,e_0 - x$ are determined by the algebra alone, so A commutes with conjugation; and $N(x)\,e_0 = x\bar{x}$, so A preserves the positive-definite norm form N. Hence $\mathrm{Aut}(\mathbb{O}, \cdot_s) \subseteq O(8)$, and it is the compact real form of G_2.

By Theorem 8 every element of $\mathrm{Aut}(\mathbb{R}^{24}, \star)$ is $h_{A,B}\rho_\pi$ with $A, B \in \mathrm{Aut}(\mathbb{O}, \cdot_s)$. By Proposition 2(5), $D \perp T$ in the standard Euclidean form of $\mathbb{R}^{24}$. On D the map $h_{A,B}$ acts as $(a, a, a) \mapsto (Aa, Aa, Aa)$, which is orthogonal for the induced form $(3N(a))$; on T it acts blockwise by B, also orthogonal; and ρ_π merely permutes the three orthogonal octonion blocks. So $h_{A,B}\rho_\pi \in O(24)$.

Finally, a $\mathbb{Z}$-linear bijection of Λ extends uniquely to $\mathbb{R}^{24}$ and its $\star$-multiplicativity extends by bilinearity, so $\mathrm{Aut}(\Lambda, +, \star) \subseteq \mathrm{Aut}(\mathbb{R}^{24}, \star) \subseteq O(24)$. Together with $g(\Lambda) = \Lambda$ this gives $\mathrm{Aut}(\Lambda, +, \star) \subseteq O(24) \cap \mathrm{GL}(\Lambda) = \mathrm{Aut}(\Lambda, +, \langle\cdot,\cdot\rangle) = \mathrm{Co}_0$. $\square$

Computational confirmation: all 36 elements were tested against the Gram matrix of the Λ-basis and all preserve $\langle u, v\rangle$ exactly; in the second pass, in standard coordinates, all 36 satisfy $g^\top g = I_{24}$ (`verify_aut_lambda_star.py`, and GAP).

Note where the weight of the argument sits. Compactness of $\mathrm{Aut}(\mathbb{R}^{24}, \star)$ is what makes $\mathrm{Aut}(\Lambda, +, \star)$ finite in the first place (Corollary 11 gives an independent finiteness proof via the lattice stabilizers), and it is what forces the isometry property. It is load bearing and cannot be replaced by the trace form.

6.1. **The trace form does not do this job.** It is tempting to argue that any $\star$-automorphism preserves the $\star$-intrinsic trace form $B(u, v) := \mathrm{tr}(L_u L_v)$, and that B is (a multiple of) the Leech form. The second half is false. Computed exactly (`verify_star_algebra_structure.py`), B is diagonal with entries $+24$ on the three e_0-coordinates and -24 on the twenty-one imaginary coordinates:

$$B(u, v) \;=\; 24 \sum_{\alpha \in \mathbb{Z}/3} \mathrm{Re}\big(u_\alpha \cdot_s v_\alpha\big), \qquad \text{signature } (3, 21).$$

It is not a scalar multiple of the Leech Gram matrix, and it is not even of the shape $\alpha\langle u_D, v_D\rangle + \beta\langle u_T, v_T\rangle$: it is blind to the D/T split and separates real from imaginary parts instead. Preservation of B therefore implies neither preservation of the Leech form nor finiteness of the automorphism group. The trace-form route fails, even though its intended conclusion is true. This was confirmed independently in the second pass.

6.2. **Relation to what the main paper (v6) asserts.** The v6 Conclusion lists as an open problem: "The automorphism group of $(\Lambda, +, \star)$, equivalently the stabilizer of the product tensor inside Co_0, and its relationship to the Conway group. Empirically, $S_3 \subseteq \mathrm{Aut}(\Lambda, +, \star) \subsetneq \mathrm{Co}_0$: the block-permutation S_3 acts by automorphisms, and the inclusion in Co_0 is strict because $-I_{24} \in \mathrm{Co}_0$ does not preserve $\star$. Full identification of $\mathrm{Aut}(\Lambda, +, \star)$ is open." Against the present calculation:

- **"Full identification is open" is now closed.** $\mathrm{Aut}(\Lambda, +, \star) \cong C_6 \times S_3$, of order 36, with the three explicit generators of §5. The open-problem list should be updated.
- $S_3 \subseteq \mathrm{Aut}(\Lambda, +, \star)$ **is correct**, and the S_3 is exactly the block-permutation subgroup. It is a proper subgroup of index 6: the blockwise C_6 was missed.
- **The containment** $\mathrm{Aut}(\Lambda, +, \star) \subseteq \mathrm{Co}_0$ **is correct and is now a theorem**, not an empirical observation. But the paper's phrasing presents it as if it were free: the phrase "equivalently the stabilizer of the product tensor inside Co_0" already presupposes that every $\star$-automorphism of Λ is an isometry. That presupposition is true, by Theorem 15, and the only proof we have of it goes through the compactness of $\mathrm{Aut}(\mathbb{O}, \cdot_s)$. If the sentence stays, it needs that justification attached.
- **The strictness** $\subsetneq \mathrm{Co}_0$ **is correct**, and the reason given ($-I_{24}$ is not multiplicative) is correct. It is a considerable understatement: 36 against $|\,\mathrm{Co}_0\,| = 8{,}315{,}553{,}613{,}086{,}720{,}000$.

7. Proved, computed, and corrected

7.1. **What is proved and what is computed.** In the table, PROOF means a mathematical argument given above; EXACT means a finite computation in exact rational or integer arithmetic which, being exhaustive on a basis of a bilinear structure or on a finite set, settles the statement outright; GAP means an independent group-theoretic identification.

Statement	Status	Where
Routing identity, Lemma 1	PROOF	§2, also EXACT
Σ is an algebra homomorphism; D, T are two-sided ideals; $D \star T = T \star D = 0$; $\mathbb{R}^{24} = D \oplus T$	PROOF	§2, also EXACT
$\varphi : (\mathbb{O}, \cdot_s) \to (D, \star)$ is an algebra isomorphism	PROOF	§2
$(D, \star)$ is simple	PROOF	§2
$(T, \star)$ is simple (multiplication algebra is all of $\mathrm{End}(T)$, dimension 256)	EXACT	§2
Every $\star$-automorphism preserves D and T; $\mathrm{Aut}(\mathbb{R}^{24}, \star) = \mathrm{Aut}(D, \star) \times \mathrm{Aut}(T, \star)$	PROOF	§2
Fourier identity $\widehat{(u \star v)}_k = \hat{u}_{2k} \cdot_s \hat{v}_{2k}$; $T \otimes \mathbb{C} \cong$ swap-twisted algebra	PROOF	§3, also EXACT over $\mathbb{Q}(\zeta)$
$\dim \mathrm{Der}(\mathbb{O}, \cdot_s) = 14$, Killing form negative definite; $\dim \mathrm{Der}(\mathbb{R}^{24}, \star) = 28$	EXACT	§4
Component group of $\mathrm{Aut}(T, \star)$ is S_3 (centralizer of the identity component has exactly 6 elements)	EXACT	§4, Route 2
$\mathrm{Aut}(\mathbb{R}^{24}, \star) = G_2 \times G_2 \times S_3$, compact, $\dim 28$	PROOF	§4
Every octonion-algebra automorphism is orthogonal; hence $\mathrm{Aut}(\mathbb{R}^{24}, \star) \subseteq O(24)$ and $\mathrm{Aut}(\Lambda, +, \star) \subseteq \mathrm{Co}_0$	PROOF	§6, also EXACT on all 36
$\Lambda \cap T = \{(p, q, r) \in (L\bar{s})^3 : p + q + r = 0\}$	PROOF	§5
$\Lambda \cap D = \{(a, a, a) : a \in Ls\}$	EXACT	§5
$[\Lambda : (\Lambda \cap D) \oplus (\Lambda \cap T)] = 3^8 = 6561$	EXACT	§5
$A(Ls) = Ls$ and $B(L\bar{s}) = L\bar{s}$ are necessary; hence $\mathrm{Aut}(\Lambda, +, \star)$ is finite	PROOF	§5
$\rho_\pi(\Lambda) = \Lambda$ for all six π	PROOF	§5
Every element of $\mathrm{Stab}(L)$ is a signed permutation of $e_1, \ldots, e_7$ fixing e_0	PROOF	§5
$|\,\mathrm{Stab}(L)| = 1344$, $|\,\mathrm{Stab}(Ls)| = 168$, $|\,\mathrm{Stab}(L\bar{s})| = 12096$	EXACT	§5

Statement	Status	Where
Structure of the three stabilizers; $12096 \cong G_2(2) = U_3(3).2$	GAP	§5
$\lvert K\rvert = 6$; every element of K has $A = B$	EXACT	§5
Diagonal part of K equals $\mathrm{Stab}(L) \cap \mathrm{Stab}(Ls) \cap \mathrm{Stab}(L\bar{s})$	PROOF	§5
$\lvert \mathrm{Aut}(\Lambda, +, \star)\rvert = 36$; structure $C_6 \times S_3$; center C_6; derived subgroup C_3; $-I_{24} \notin G$	EXACT + GAP	§5
Trace form $\mathrm{tr}(L_u L_v) = 24 \sum_\alpha \mathrm{Re}(u_\alpha \cdot_s v_\alpha)$, signature $(3, 21)$, not the Leech form	EXACT	§6

Nothing load bearing is left open. The one step that is neither an exact finite computation nor elementary is Route 1 of §4(c), the classification of $\mathrm{Aut}(S)$ over $\mathbb{C}$; see Remark 9. It is not load bearing, because Route 2 reaches the same conclusion by an entirely finite and machine-checkable path.

7.2. **Corrections made during verification.** Five points where the first computational pass, or the expectation that launched it, had to be corrected. All are recorded here rather than quietly fixed.

- **C1. The intrinsic-annihilator argument is circular as first stated** (Remark 6). The conclusion survives; the proof is now via simplicity of the two ideals and a dimension count. The docstring of `verify_star_algebra_structure.py` still carries the circular wording.
- **C2. The order-12096 group stabilizes $L\bar{s}$, not L** (Remark 12). The prior expectation that the maximal-order stabilizer would be $G_2(2)$, cut down by the extra sublattice conditions, is wrong in the paper's normalization.
- **C3. The trace-form route to Co_0 fails** (§6). Its conclusion is true, but for the compactness reason instead.
- **C4. The element-order list first reported was the list of conjugacy class representative orders**, not of group elements.
- **C5. Scale.** The Λ realized by (1) has minimal norm 8 and Gram determinant 2^{24}: it is $\sqrt{2}$ times the unimodular Leech lattice. This is irrelevant to Aut but no unimodularity claim should be attached to it.

8. Reproduction

All scripts use exact arithmetic (Python `Fractions` or integers) throughout; no claim in this note rests on floating-point computation. Runtimes are on the project's reference machine. Python scripts live in `python_project/src/` and GAP scripts in ; both are named below by file name alone.

Script	Certifies
`verify_ideal_decomposition.py`	Σ is an algebra homomorphism; $\mathbb{R}^{24} = D \oplus T$ as mutually annihilating two-sided ideals; $(\mathbb{O}, \cdot_s) \cong (D, \star)$; the twist is an algebra isomorphism onto the untwisted triple product.
`verify_star_algebra_structure.py`	The routing identity; $\mathrm{Ann}(T) = D$ and $\mathrm{Ann}(D) = T$; the C_3-Fourier normal form checked exactly over $\mathbb{Q}(\zeta)$ on all 576 basis pairs; $h_{A,B}$ and the block permutations are automorphisms; $\dim \mathrm{Der}(\mathbb{O}, \cdot_s) = 14$ and $\dim \mathrm{Der}(\mathbb{R}^{24}, \star) = 28$; the $\star$-intrinsic trace form. About 2 minutes.

Script	Certifies
`verify_aut_lambda_star.py`	The main script. $\Lambda \cap D$ and $\Lambda \cap T$; complete enumeration of the three octonion-automorphism stabilizers; the pair set K by the two independent methods M1 and M2; the 36 group elements verified one at a time against Λ, against $\star$ on all 576 basis pairs, and against the Leech Gram matrix; writes the GAP generators. About 5 minutes.
`verify_aut_octonion_crosscheck.py`	Independent cross-checks of the three stabilizer enumerations: re-run from a different generating triple, and an assumption-free brute force over all 645,120 signed permutations for $\mathrm{Stab}(L)$; writes 8×8 GAP generators. About 9 minutes.
`aut_lambda_star.g`	Reads three explicit 24×24 integer matrices and reports Size = 36, StructureDescription = `C6 x S3`, center C_6, derived subgroup C_3, and $-I_{24} \notin G$.
`octonion_stabilisers.g`	Identifies $\mathrm{Stab}(L) = (C_2{\times}C_2{\times}C_2).\mathrm{PSL}(3,2)$ of order 1344; $\mathrm{Stab}(Ls) = (C_2{\times}C_2{\times}C_2){:}(C_7{:}C_3)$ of order 168; $\mathrm{Stab}(L\bar{s}) = \mathrm{PSU}(3,3){:}C_2$ of order 12096, confirmed isomorphic to the ATLAS group $U_3(3).2 = G_2(2)$.
`aut_lambda_star_gens.g`, `octonion_stabilisers_gens.g`	Machine-generated generator files. Do not edit.

The earlier probe `python_project/src/probe_aut_lambda_star.py` recorded the S_3 of block permutations and the exclusion of $-I_{24}$; those observations are subsumed here.

The independent second pass rebuilt the octonions, σ, $\cdot_s$, L, s, $\bar{s}$, Ls, $L\bar{s}$ and a $\mathbb{Z}$-basis of Λ from the definitions in §1 (the Λ-basis from the defining congruences, by an $\mathbb{F}_2$-linear solve followed by Hermite reduction, rather than from a generating set), rebuilt the three generators in standard coordinates from their intrinsic description, and re-derived the ambient group by Route 2. It used none of the scripts above. Its working files are not part of the repository.

Anthropic